Parson and Sheingorn [PS1], [PS2]. Interactions between function theory, covering spaces and amenability also appear in papers of Brooks [Br] and Lyons and Sullivan [LS].

We have recently learned that Ohtake proved $\|\Theta\| = 1$ for abelian coverings [Oh].

2. Nonamenable graphs and slowly varying functions

In this section we will isolate some combinatorial properties of nonamenable groups, and more generally nonamenable *graphs*. The added generality will allow us to deal with irregular covers.

Definitions. Let G be a graph. For any set V of vertices of G, define $b(V)$, the *border* of V, to be the set of vertices at distance 1 from V. (A vertex at distance 1 is connected to V by an edge but does not itself lie in V).

Define the *expansion* γ of G as the infimum of $|b(V)|/|V|$ as V ranges over all finite sets of vertices. If the expansion is 0, G is *amenable*. Otherwise G is *nonamenable*, and the size of the border of any set of vertices is a definite fraction of the size of the set itself.

For example: an infinite tree in which every vertex has degree d is amenable for $d=2$, but nonamenable, with expansion $\gamma=(d-2)$, when d is 3 or more.

Now let f: $V \to \mathbf{R}$ be a positive integrable function on a subset V of the vertices of G (i.e. $\sum_{x \in V} f(x) < \infty$.) For x in the border $b(V)$, define

$$F(x) = \sup\{f(y): y \text{ is adjacent to } x.\}$$

Theorem 2.1. *Suppose that G is nonamenable with expansion γ, and f is slowly varying in the sense that, for some λ, $0 < \lambda < 1$,*

$$f(x) \geqq \lambda f(y)$$

whenever x and y are adjacent vertices of V. Then

$$\sum_{x \in b(V)} F(x) \geqq (\lambda\gamma + \lambda - 1) \sum_{x \in V} f(x).$$

Remarks.

1. Of course $f(x) \leqq \lambda^{-1} f(y)$ by symmetry.
2. If f varies sufficiently slowly, a definite fraction of its mass is picked up by F. This mass will give a definite contribution to the contraction of Θ in § 7.
3. Setting $f \equiv 1$ on V and $\lambda = 1$ gives $|b(V)| \geqq \gamma |V|$ as a special case.

Proof. For every integer n, let V_n denote the subset of V on which $f(x) \geqq \lambda^n$. Then V_n is an increasing sequence of finite sets and V_n is empty for n sufficiently negative.

Let $B_n = b(V_n) \cap b(V)$, and let v_n and b_n denote $|V_n|$ and $|B_n|$ respectively. By the slow variation of f, $b(V_n) \subset B_n \sqcup (V_{n+1} - V_n)$, so by nonamenability,

$$b_n + \Delta v_n \geqq \gamma v_n,$$

where $\Delta v_n = v_{n+1} - v_n$ is the usual combinatorial derivative. Applying this inequality after integrating by parts (note that $\Delta \lambda^n < 0$), we find

$$\sum \lambda^{n+1} \Delta b_n = -\sum b_n \Delta \lambda^n \geqq \sum \Delta v_n \Delta \lambda^n - \gamma v_n \Delta \lambda^n,$$

where we sum over all integers n. Integrating the second term on the right by parts, this can be rewritten

$$\sum \lambda^{n+1} \Delta b_n \geqq (\lambda - 1 + \lambda\gamma) \sum \lambda^n \Delta v_n.$$

Now $f(x)$ is bounded above by λ^n on $V_{n+1} - V_n$, so the sum appearing on the right is an upper bound for $\sum f(x)$; similarly the left hand sum is a lower bound for $\sum F(x)$, establishing the theorem. □

In geometric situations, we will deal exclusively with graphs of bounded degree (there is a bound on the number of edges attached to each vertex). A set of vertices I is *independent* if no two elements of I are joined by an edge.

Proposition 2.2. *Let G be a graph of degree at most d, f: $V \to \mathbf{R}$ a non-negative integrable function on the vertices of G. Then there exists a set of independent vertices I such that*

$$\sum_I f(x) \geqq \frac{1}{d+1} \sum_V f(x).$$

Proof. Sort the vertices of G so $f(x)$ is descending ($f(x_0) \geqq f(x_1) \geqq \ldots$), and construct I by a greedy algorithm: recursively add to I the next vertex in sequence which is not adjacent to those already chosen. Then every vertex y not in I can be assigned to an adjacent vertex x in I so $f(x) \geqq f(y)$. This assignment is at most d-to-1, and the proposition follows. □

3. From coverings to graphs

In this section we define (in several equivalent ways) the notion of an amenable, but possibly irregular covering. ([Gre] is a concise reference for the theory of amenable groups, of which this material is a generalization; see also [Pier].)

All manifolds considered are assumed smooth and connected. Let X be a manifold, p: $Y \to X$ a covering, p_*: $\pi_1(Y) \subset \pi_1(X)$ the induced inclusion of fundamental groups (for our considerations basepoints will be unimportant).

Definitions. The covering $Y \to X$ is *amenable* if there is a linear functional

$$m\colon L^\infty(\pi_1(X)/\pi_1(Y)) \to \mathbf{R},$$

invariant under the left-action of $\pi_1(X)$, such that $\inf(f) \leqq m(f) \leqq \sup(f)$ for all f.

Such a functional m is an invariant mean for the group action.

A *group* G is amenable if $L^\infty(G)$ admits a G-invariant mean. Thus, amenability of a regular covering is equivalent to amenability of its group of deck transformations.

Examples.

1. Solvable groups are amenable, as are subgroups and quotient groups of amenable groups.

2. The free group on 2 or more generators is nonamenable.

3. Any nonelementary Fuchsian group contains a free group on 2 generators, and hence is nonamenable.

4. In particular, the fundamental group of hyperbolic Riemann surface of finite type is nonamenable.

5. Subexponential growth implies amenability, but there are amenable groups with exponential growth – such as the solvable group $\langle x, y\colon xy = y^2 x\rangle$.

Using graphs, one can give a more combinatorial characterization of an amenable covering.

Assume for the moment that $\pi_1(X)$ has a finite generating set $\mathcal{G}$. From this data, construct the *coset graph* as follows: take the cosets $\pi_1(X)/\pi_1(Y)$ as vertices, and connect $x\pi_1(Y)$ and $y\pi_1(Y)$ by an edge if there exists a g in $\mathcal{G}$ such that $gx\pi_1(Y) = y\pi_1(Y)$. This graph is combinatorial model for the covering.

It will be useful to introduce finer models, as follows. A *net* $\mathcal{E}$ on X is a collection of simply connected open sets, whose union U is connected, such that $E \cap E'$ is either empty or connected for any two members of $\mathcal{E}$.

To $\mathcal{E}$ we associate its *Čech graph*, whose vertices are the elements of $\mathcal{E}$ and whose edges join E and E' iff $E \cap E' \neq \emptyset$. The Čech graph is the same as the 1-skeleton of the nerve of $\mathcal{E}$.

There is an induced net on Y, which we denote by $p^*(\mathcal{E})$, whose elements are the *components* of $p^{-1}(E)$ for each E in $\mathcal{E}$.

Our second combinatorial model for the covering $p\colon Y \to X$ is the Čech graph of $p^*\mathcal{E}$.

Proposition 3.1. *A covering $p\colon Y \to X$ is amenable iff one of the following equivalent conditions is satisfied:*

1. *The coset graph of $K/(K \cap \pi_1(Y)$ is amenable for every finitely generated subgroup K of $\pi_1(X)$.*

2. *The pullback of the covering to X_0 is amenable for every submanifold of X_0 of X with finitely generated fundamental group.*

3. *The Čech graph of $p^*\mathcal{E}$ is amenable for every finite net $\mathcal{E}$ on X.*

Equivalence of the three conditions is easy to check. That the first condition is equivalent to the existence of an invariant mean on the coset space is essentially Rosenblatt's generalization of Følner's condition [Ros].

As we will never make use of the invariant mean, the reader may also take these combinatorial conditions as the definition of amenability.

4. Explicit expansion constants

Let $Y \to X$ be a covering of surfaces, $H \hookrightarrow G$ the corresponding inclusion of fundamental groups. Assume both H and G are finitely generated. We will give an estimate for expansion constant γ of the coset graph G/H which depends only on Euler characteristics.

This estimate provides many examples of *irregular* nonamenable coverings, and will also be used to obtain bounds on $\|\Theta\|$ which depend only on the location of X in moduli space (§ 10).

Of course the expansion constant depends on a choice of generating set $\mathcal{G}$ for G. Any surface group may be presented as:

$$G = \langle a_1, \ldots, a_n \rangle \text{ or}$$
$$G = \langle a_1, b_1, \ldots, a_g, b_g : [a_1, b_1] \ldots [a_g, b_g] = 1 \rangle$$

and we will use these generators for $\mathcal{G}$.

The Euler characteristics of G and H are the same as the topological Euler characteristics of X and Y. Assume G is nontrivial; then $\chi(G) \leqq 0$.

Lemma 4.1. *For any finite set of vertices $V \neq \emptyset$ in the coset graph G/H,*

$$|b(V)| \geqq 2(\chi(H) - |V|\,\chi(G)).$$

Proof. The presentation given for G corresponds to a cell complex C which is either a bouquet of circles or a compact surface. There is a covering $\tilde{C}$ corresponding to H, whose 1-skeleton is the coset graph.

Let $V \subset \tilde{C}$ be a finite set of vertices, and let D be the largest subcomplex of $\tilde{C}$ whose 0-skeleton is $V \cup b(V)$. For the purposes of the estimate, we may assume no component of $\tilde{C} - D$ is a 2-disk, since otherwise we may include its vertices in V and increase $|V|$ while decreasing $b(V)$. With this assumption, $\chi(D) \geqq \chi(\tilde{C}) = \chi(H)$.

Let C_i, D_i denote the number of i-cells in C and D. Noting that the degree of the coset graph (the number of edges incident to each vertex) is $2C_1$, we find:

$$D_0 = |V| + |b(V)| = C_0|V| + |b(V)|$$
$$D_1 = \tfrac{1}{2}\sum_{D_0} \deg(v) \geqq \tfrac{1}{2}(2C_1|V| + |b(V)|)$$
$$D_2 \leqq |V|\,C_2$$

from which the theorem follows. □

Corollary 4.2 (Uniform expansion). *Assume $[G:H] = \infty$. Then the expansion of the coset graph satisfies*

$$\gamma \geqq \begin{cases} 2|\chi(G)| & \text{if } \chi(H) \geqq 0, \\ |\chi(G)/\chi(H)| & \text{otherwise.} \end{cases}$$

In particular, if $\chi(G) < 0$ the coset graph is nonamenable with expansion bounded below independent of the embedding of H into G.

Proof. Combine the preceding Theorem with the estimate $b(V) \geqq 1$, which holds so long as the coset graph is infinite. □

Example. Let X be a hyperbolic Riemann surface of finite type, and let $S \subset X$ be a proper incompressible subsurface. (*Incompressible* means $\pi_1(S)$ injects into $\pi_1(X)$; *proper* means the image is a proper subgroup.) Corresponding to the subgroup $\pi_1(S)$ there is an infinite-sheeted covering $Y \to X$, which is nonamenable

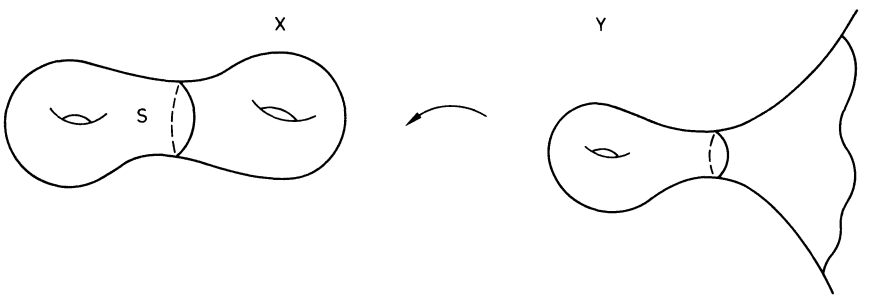


Fig. 1. The covering space of a subsurface

by the result above. Furthermore the expansion is bounded below independent of the choice of S, since $\chi(X) < \chi(S) \leqq 0$.

Coverings induced by subsurfaces are said to be *geometric*. Any covering $Y \to X$ with $\pi_1(Y)$ finitely generated is virtually geometric, i.e. it becomes geometric when pulled back to a finite covering of X [Scott].

5. Flex of the unit sphere in $Q(\Delta)$

This section lays the analytic groundwork for the proof of Theorem 1.1.

Let $Q(X)$ be the Banach space of quadratic differentials on a Riemann surface X, introduced in the § 1. There is a natural pairing between $Q(X)$ and $M(X)$, the space of measurable Beltrami differentials $\mu(z)\, d\bar{z}/dz$ with

$$\|\mu\| = \sup_X |\mu(z)| < \infty,$$

namely

$$\langle \phi, \mu \rangle = \mathrm{Re} \int_X \phi(z)\, \mu(z)\, |dz|^2.$$

In fact the dual of $Q(X)$ is the quotient of $M(X)$ by the subspace annihilated by Q.

Given any ϕ of norm 1, $\mu = \bar{\phi}/|\phi|$ is the unique element of norm 1 in $M(X)$ such that $\langle \phi, \mu \rangle = 1$. Geometrically, this means the unit sphere in $Q(X)$ has no sharp corners – at each point there is a unique supporting hyperplane.

On the unit disk Δ, consider the quadratic differential $\psi(z) = dz^2/\pi$. Then ψ is a point on the unit sphere in $Q(\Delta)$.

Theorem 5.1. *Let ϕ be an element of the unit sphere in $Q(\Delta)$, such that*

$$\langle \phi, \bar{\psi}/|\psi| \rangle = 1 - \varepsilon.$$

Then $\|\psi - \phi\| < O(\varepsilon^\alpha)$ for some fixed exponent α.

Remark. The pairing between ϕ and $\bar{\psi}/|\psi|$ gives the average, with respect to $|\phi|$-measure, of the cosine of twice the angle between the foliations of ϕ and ψ. (The foliation of ψ is just the foliation of the disk by horizontal lines.) The theorem says that if the foliations are nearly synchronized, then ϕ is close to ψ.

Proof. Writing $\|\cdot\|$ and $\langle , \rangle$ in coordinates, our task is to show that

$$\int_\Delta |\phi(z)| = 1$$

and

$$\int_\Delta \operatorname{Re} \phi(z) = 1 - \varepsilon$$

together imply

$$\int_\Delta |1 - \phi(z)| = O(\varepsilon^\alpha),$$

all integrals being with respect to normalized area measure on the disk.

In any measure space, the first two equations imply

$$\int_\Delta |\operatorname{Im} \phi(z)| = O(\varepsilon^{1/2}).$$

Since $\operatorname{Im} \phi(z)$ is harmonic, it is controlled by its average over a ball and we obtain

$$|\operatorname{Im} \phi(z)| = O(\varepsilon^{1/2}/d(z)^2),$$

where $d(z) = d(z, \partial\Delta)$ denotes Euclidean distance from z to the boundary of the disk. Thus ϕ maps $B(0, 1-d)$ into the strip $\{z : |\operatorname{Im} z| < O(\varepsilon^{1/2}/d^2)\}$.

Similarly $\operatorname{Re} \phi$ is harmonic, so its average over the disk gives the value at the origin: $\operatorname{Re} \phi(0) = 1 - \varepsilon$. By the Schwarz lemma, most of $B(0, 1-d)$ maps near $\phi(0)$, since the image of $B(0, 1-d/2)$ is confined to a narrow strip. Making this precise, one obtains

$$|1 - \phi(z)| = O(\varepsilon + \varepsilon^{1/2} \log(1/d(z))/d(z)^2).$$

Finally we control $|\phi|$ near the boundary of the disk: since $|\phi|$ is subharmonic,

$$1 - \varepsilon \leqq |\phi(0)| \leqq \frac{1}{(1-d)^2} \int_{B(0,\,1-d)} |\phi|$$

and since the total integral of $|\phi|$ is 1, this implies

$$\int_{\{1-d<|z|<1\}} |\phi| = O(d + \varepsilon).$$

Combining this with the uniform bound for $|1-\phi(z)|$ on $B(0, 1-d)$, one obtains

$$\int_{\Delta} |1-\phi| = O(\varepsilon + d + \varepsilon^{1/2} \log(1/d)/d^2) = O(\varepsilon^{1/6} \log(1/\varepsilon))$$

if we set $d=\varepsilon^{1/6}$. Thus the theorem is true for any exponent $\alpha < 1/6$. □

Remarks.

1. In terms of the geometry of Banach spaces, the theorem states that a point of the unit sphere in $Q(X)$ which is close to the supporting hyperplane for ψ is close to ψ itself; intuitively, the sphere is 'flexed' at ψ. If $Q(X)$ were a Hilpert space with its usual norm, the theorem would be true with $\alpha = 1/2$.

2. Conversely, say a point p of the unit sphere S of a Banach space is *flat* if for any hyperplane H supporting S at p, there exist $\langle p_n \rangle$ of norm 1 whose distance from H tends to 0 but $\|p-p_n\| > \varepsilon > 0$. Then many points of the unit sphere $Q(X)$ are flat.

For example, let $\langle z_i \rangle$ be a sequence of points in the disk such that the hyperbolic balls $B(z_i, r_i)$ are disjoint and $r_i \to \infty$. Let M_i: $\Delta \to \Delta$ be a sequence of Möbius transformations such that $M_i(z_i) = 0$, and consider the differential

$$\alpha = c \sum_{i=1}^{\infty} 2^{-n} M_i^*(dz)^2,$$

where the constant c is chosen to give this expression norm in 1 in $Q(\Delta)$. It is not hard to check that $\langle M_i^*(dz^2/\pi), \bar{\alpha}/|\alpha| \rangle \to 1$ but these differentials do not approach α in the norm topology.

3. There is a close relationship between flexing of the unit sphere and reflexivity. For example, a theorem of Milman asserts that a uniformly convex Banach space is reflexive [Mil]; see also [Pet]. $Q(\Delta)$ is not reflexive, but its unit ball *is* uniformly convex at certain points, such as ψ.

6. The nonamenable case: selecting the net

Fix the following data:

- A Riemann surface X;
- A quadratic differential ψ in $Q(X)$ with $\|\psi\| = 1$; and
- A nonamenable covering p: $Y \to X$, with induced relative Poincaré operator

$$\Theta\colon\ Q(Y) \to Q(X).$$

For this data, we will establish the following 'fiberwise' version of the Theta Conjecture:

Theorem 6.1 (Θ contracts fibers). *Let $Q_\psi(Y)$ denote the closed subspace of $Q(Y)$ consisting of differentials ϕ such that $\Theta(\phi)$ is a multiple of ψ. Then $\|\Theta | Q_\psi(Y)\| < 1$.*

To set up the proof, we first construct a net $\mathscr{E}$ on X such that $p^*\mathscr{E}$ detects the nonamenability of the covering.

Recall that ψ gives rises to a flat conformal metric on X with quadratic form $|\psi(z)|\,|dz|^2$ (see e.g. [Str3]). At the zeros of ψ the metric has cone-like singularities; these will not concern us.

If $\psi(x)\neq 0$, there is a neighborhood V of x and a conformal isomorphism $g\colon (V,0)\to(\Delta_r,0)$ such that $\psi|V=g^*(dz^2)$ (here Δ_r denotes the disk of radius r centered at the origin in $\mathbf{C}$). The map g is an isometry from the ψ-metric to the Euclidean metric, and V is an *embedded* r-ball in the ψ-metric.

Let $r(x)$ denote the *injectivity radius* of ψ at x, i.e. the radius of the largest embedded r-ball centered at x.

Now choose a finite net $\mathscr{E}$ on X such that, for some fixed $r>0$,

(a) Each element of $\mathscr{E}$ is an embedded r-ball in the ψ-metric; and

(b) The Čech graph of $p^*\mathscr{E}$ is nonamenable.

This is easily done, for example as follows. By nonamenability, there is a finitely generated subgroup K of $\pi_1(X)$ whose coset graph is nonamenable. Choose a basepoint at which $\psi\neq 0$ and smooth loops generating this subgroup. The loops may also be chosen to avoid the zeros of ψ. By compactness, the injectivity radius of ψ on the loops is bounded below by some $s>0$. Let $\mathscr{E}$ be a finite covering of the union of the loops by embedded r-balls, where $r=s/2$. Any two balls which meet are contained in an embedded s-ball, so their intersection is connected, just as in Euclidean space; thus $\mathscr{E}$ is a net. Finally the subgroup K is carried by $\mathscr{E}$, so the Čech graph of $p^*\mathscr{E}$ is nonamenable.

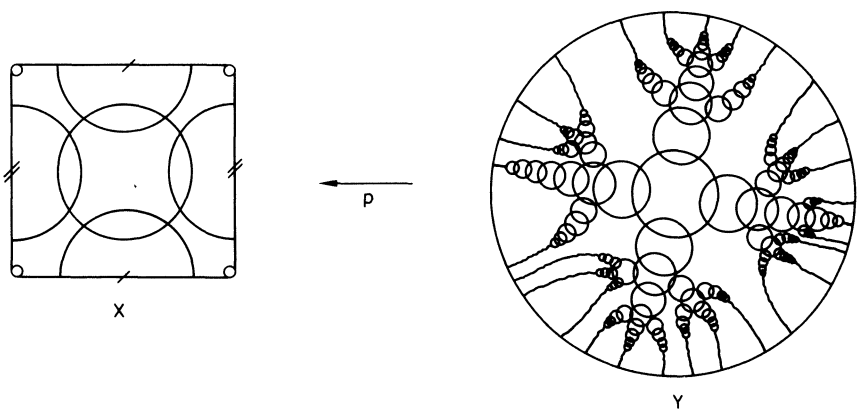


Fig. 2. A net for the punctured torus

For an example in the case of the universal covering of a punctured torus, see Fig. 2. Our bound on $\|\Theta\|$ will depend on the following quantities:

1. The expansion constant of the Čech graph of the induced net $\mathscr{F}=p^*\mathscr{E}$;
2. The degree of this graph (which is at most $|\mathscr{E}|$);
3. The ψ-area of $\bigcup\mathscr{E}$; and
4. The *minimum overlap* between elements of $\mathscr{E}$. This is the minimum fraction of the ψ-area of an element E which is shared with an overlapping element E'.

7. Capturing $\|\Theta\|$ fiberwise

Let ϕ be an element of $Q_\psi(Y)$ such that $\|\phi\|=1$. To establish Theorem 6.1, we will bound $\|\Theta(\phi)\|$ away from 1 by an amount which is independent of ϕ. In this section, any bounds written using $O(\cdot)$ will involve constants which are independent of ϕ.

$\Theta(\phi)=t\psi$ for some constant t. Replacing ϕ by $t\phi/|t|$ if necessary, we many assume $t\geqq 0$.

Let $\mu=\bar{\psi}/|\psi|$ denote the unique element of norm 1 in $M(X)$ such that $\langle\psi,\mu\rangle=1$. For F a measurable subset of Y, define the *mass* and *inefficiency* of F by:

$$m(F)=\int_F |\phi|,$$

$$i(F)=1-\frac{1}{m(F)}\operatorname{Re}\int_F \phi\, p^*\mu.$$

Since $m(Y)=1$,

$$1-i(Y)=\langle\phi, p^*\mu\rangle=\langle\Theta(\phi),\mu\rangle=\langle t\psi,\mu\rangle=t,$$

so $i(Y)=1-\|\Theta(\phi)\|$. Our goal is to bound the inefficiency of Y from below.

Mass and inefficiency are always non-negative, so

$$i(Y)=m(W)\,i(W)+m(Y-W)\,i(Y-W)\geqq m(W)\,i(W),$$

and hence we need only find a set $W\subset Y$ which has definite mass and inefficiency.

Lemma 7.1. *For any F in $\mathcal{F}$, there is a constant c such that*

$$\int_F |c\phi-p^*\psi|\leqq O(i(F)^\alpha)$$

for some fixed exponent α.

Proof. Choose a coordinate z so that F can be identified with the unit disk Δ. After an appropriate rotation, $p^*\psi=r^2\,dz^2$ so $p^*\mu=d\bar{z}/dz$. By Theorem 5.1 on the flex of the sphere in $Q(\Delta)$,

$$\int_F \left|\frac{\phi}{m(F)}-\frac{dz^2}{\pi}\right|\leqq O(i(F)^\alpha),$$

and multiplying through by πr^2 we obtain the Lemma. □

Corollary 7.2. *For all intersecting F and F' in $\mathcal{F}$, and for all i_0 sufficiently small (independent of ϕ),*

1. *If the inefficiency $i(F)$ is less than i_0, then $m(F')\geqq O(m(F))$;*
2. *If $i(F)$ and $i(F')$ are both less than i_0, then $m(F')\geqq\lambda m(F)$, where $\lambda=1-O(i_0^\alpha)$.*

In other words, efficient neighbors have nearly the same mass, and a definite amount of mass is inherited from an efficient neighbor.

Proof. Assume both inefficiencies are near 0. By the Lemma above, there are constants c and c' such that $c\phi$ and $c'\phi$ are L^1-close to $p^*\psi$ on F and F' respectively. Thus $m(F)$ and $m(F')$ are given approximately by $\pi r^2/c$ and $\pi r^2/c'$.

Now F and F' overlap a definite amount with respect to $p^*\psi$-area, since their overlap is bounded below by the minimum overlap in $\mathscr{E}$. It follows that the constants c and c' must be close, else there would be a definite deviation from $p^*\psi$ on the overlap. Making this precise yields the bound given.

If only $i(F)$ is near zero, it still follows that the ϕ-mass of the overlap is a definite fraction of $m(F)$. The overlapping mass also contributes to $m(F')$, so $m(F') \geqq O(m(F))$. □

Lemma 7.3. *For any subset $\mathscr{F}'$ of $\mathscr{F}$,*

$$m(\bigcup \mathscr{F}') \geqq \frac{1}{|\mathscr{E}|} \sum_{\mathscr{F}'} m(F)$$

and

$$i(\bigcup \mathscr{F}') \geqq \frac{\inf\{i(F): F \in \mathscr{F}'\}}{|\mathscr{E}|}.$$

Proof. The degree of the Čech graph of $\mathscr{F}$ is at most $|\mathscr{E}|-1$, so Proposition 2.2 applies to give an independent subset $\mathscr{F}'' \subset \mathscr{F}'$ which carries at least $1/|\mathscr{E}|$ the sum of $m(\cdot)$ over $\mathscr{F}'$. Then $\bigcup \mathscr{F}''$ is a disjoint union (this is what independence means), so the mass is additive, establishing the first inequality.

By disjointness, $i(\bigcup \mathscr{F}'')$ is bounded below by the minimum inefficiency of members of $\mathscr{F}'$. Estimating $i(\bigcup \mathscr{F}')$ by the weighted average of the inefficiency of $\bigcup \mathscr{F}''$ and its complement gives the second inequality. □

Proof of Theorem 6.1 (Θ contracts fibers). Let $\gamma > 0$ denote the expansion of the (nonamenable) Čech graph of $\mathscr{F}$. Choose $\lambda < 1$ such that $\lambda\gamma + \lambda - 1 \geqq \gamma/2$. Let

$$\mathscr{V} = \{F \in \mathscr{F} : i(F) < i_0\},$$

where $i_0 > 0$ is chosen sufficiently small that we may apply Corollary 7.2 to conclude:

(I) $m(F') \geqq O(m(F))$ whenever F' is adjacent to an element $F \in \mathscr{V}$; and

(II) $m(F') \geqq \lambda m(F)$ whenever F and F' are adjacent elements of $\mathscr{V}$.

Let

$$\mathscr{W} = \{F \in \mathscr{F} : i(F) \geqq i_0\}.$$

Then $(\mathscr{V}, \mathscr{W})$ forms a partition of $\mathscr{F}$. Let $W = \bigcup \mathscr{W}$, $V = \bigcup \mathscr{V}$.

By Lemma 7.3, $i(W) \geqq i_0/|\mathscr{E}|$, so

$$1 - \|\Theta(\phi)\| = i(Y) \geqq m(W)\, i(W) \geqq i_0\, m(W)/|\mathscr{E}|$$

and hence to bound $\|\Theta(\phi)\|$ away from 1, it suffices to bound $m(W)$ from below.

Assume $\|\Theta(\phi)\| \geqq 1/2$, since otherwise we are done. Then

$$m(W)+m(V)=\int_{p^{-1}(\bigcup \mathscr{E})} |\phi| \geqq \|\Theta(\phi)\| \int_{\bigcup \mathscr{E}} |\psi| = O(1),$$

so to complete the proof it suffices to show the mass of W is not too much smaller than the mass of V.

For F in the border $b(\mathscr{V}) \subset \mathscr{W}$, define

$$M(F)=\sup\{m(F'): F' \in \mathscr{V} \text{ and } F' \text{ is adjacent to } F.\}$$

By (I), $m(F) \geqq O(M(F))$, so by Lemma 7.3 the mass of W is estimated from below in terms of the sum of $M(F)$ over the border of $\mathscr{V}$. We will show this sum is comparable to the mass of V.

By (II), $m(\cdot)$ is a slowly varying function on the vertices $\mathscr{V}$ of the Čech graph; applying Theorem 2.1,

$$\sum_{F \in b(\mathscr{V})} M(F) \geqq (\lambda\gamma+\lambda-1) \sum_{F \in \mathscr{V}} m(F) \geqq \frac{\gamma}{2} m(V),$$

establishing the theorem. □

8. Completion of the nonamenable case

It is known that the operator Θ is always surjective; this fundamental property is called *completeness of Poincaré series*. By the open mapping theorem, the image of the unit ball in $Q(Y)$ contains a neighborhood of the origin in $Q(X)$. This can be made effective:

Theorem 8.1. *Let $Y \to X$ be a covering of a hyperbolic Riemann surface X. Then*

$$\Theta(B_Y) \supset \tfrac{1}{3} B_X,$$

i.e. the image of the unit ball in $Q(Y)$ contains the ball of radius 1/3 in $Q(X)$.

This result is due to Ahlfors and Bers; for a proof, see [Kra 1, Theorem 3.3], or [Gar, § 4.3].

Proof of Theorem 1.1 (nonamenable case). Let $Y \to X$ be a nonamenable covering of a hyperbolic Riemann surface X. Suppose ψ is an element of norm 1 in $\overline{\Theta(B_Y)}$. Then there is a sequence ϕ_n in B_Y such that $\Theta(\phi_n) \to \psi$. By the Theorem above, there exist $\phi_n' \to 0$ with $\Theta(\phi_n')=\psi-\Theta(\phi_n)$. Then $\Theta(\phi_n+\phi_n')=\psi$ and $\|\phi_n+\phi_n'\| \to 1$, contradicting Theorem 6.1 (Θ contracts fibers).

Therefore the closure of $\Theta(B_Y)$ is contained in the interior of the unit ball in $Q(X)$. □